

The Anachronistic Testing-Date Bias of Machine Learning Strategies

Federico Tiberger

Management Science and Operations, London Business School,
London, United Kingdom, NW1 4SA, ftiberger@london.edu

We examine whether the documented outperformance of machine learning models in cross-sectional stock return prediction reflects genuine economic alpha or is partly an artefact of an anachronistic testing-date bias, that is, the systematic back-testing of models on periods before their public availability, when markets had no opportunity to arbitrage the strategy. We study eight predictive models: Elastic Net, XGBoost, LightGBM, CatBoost, and four feedforward neural networks of increasing depth over the out-of-sample period from January 2005 to December 2024. We form value-weighted long-short decile portfolios and evaluate their performance gross and net of transaction costs. Gross long-short returns are positive and significant for all models, except XGBoost and LightGBM. Once transaction costs are deducted, none of the strategies delivers a statistically significant positive net return. We then examine whether the predictive power of models declines after their public release by comparing returns before and after each model's public release. We find that post-release returns and alphas are generally lower than pre-release levels. The strongest statistical evidence comes from a pooled regression that estimates an average post-release decline in returns of 0.512% per month, statistically significant at the 5% level. Factor-adjusted regressions also yield negative estimates, though with weaker statistical support. These findings suggest that a non-trivial share of the machine-learning return premium documented in the literature reflects a technological advantage that erodes as the underlying methods become widely adopted.

Key words: Machine Learning, Cross-Sectional Asset Pricing, Return Predictability, Transaction Costs, Publication Decay, Limits of Arbitrage, Market Efficiency

1. Introduction

Following the groundbreaking research by Gu et al. (2020), a significant strand of financial research has explored machine learning's ability to predict the cross-section of stock returns (Gu et al. 2021, Chen et al. 2024, Azevedo et al. 2023). To evaluate their performance, researchers commonly

use backtesting methods that assume full access to data, features, and architectures throughout the entire sample period. This approach implicitly allows researchers to assess an algorithm during time periods predating its actual invention, large-scale feasibility, or adoption into financial markets. This practice introduces what we define as anachronistic testing-date bias. Specifically, anachronistic testing-date bias occurs when the information set $\mathcal{I}_t$ used at time t in the training and test sets includes data transformations, predictors, or model architectures that were not economically or technologically feasible at time t . We can further decompose this bias into three categories: *signal availability*, which occurs when we use predictors before they are discovered, as discussed by McLean and Pontiff (2016), *model feasibility*, which occurs when we train architectures before they are invented or computationally feasible, and *market awareness*, which happens when a signal or architecture exists, but markets were still unaware of it. For example, although neural networks were developed over half a century ago, they only gained widespread recognition in the last 15 years. In this paper, we study the effects of model feasibility on returns. A useful metric for measuring the performance of a financial strategy is the alpha, defined as the intercept in the regression

$$R_{i,t} = \alpha_i + \beta_i^\top f_t + \varepsilon_{i,t},$$

where f_t denotes a vector of benchmark risk factor returns, and β_i represents the corresponding exposure of strategy i to those factors. Essentially, alpha measures the risk-adjusted performance of the strategy: given the level of risk it takes, how well does it perform? For instance, a common benchmark risk factor is the *market return* MKT. Let's consider a MKT single-factor specification and a strategy with $\beta_{\text{MKT}} = 1$, meaning we expect it to co-move one-for-one with the market. If, over a given period, the entire market returns 20% while the strategy returns only 15%, we cannot claim the strategy performed well during that period, even though it was still profitable. But if alpha depends on market behaviour, can one meaningfully compare how their new model would have performed in a market unaware of its existence? This paper argues that such comparisons systematically overstate expected returns and proposes a framework to restore temporal and

economic consistency.

The literature has shown that using information not yet discovered during the testing periods can lead to overestimating a model’s effectiveness. McLean and Pontiff (2016) show that trading strategies and anomalies lose 32% in returns after being published. Similarly, Azevedo et al. (2023) show an average cumulative performance reduction of 57% when accounting for transaction costs and post-publication decay. However, although studies such as Azevedo et al. (2023) treat the predictive set as time-varying, they typically assume that the technology used to generate predictions is constant and readily available throughout the sample period. This assumption is unlikely to be accurate in practice, as many machine learning methods depend on advances in optimisation algorithms, software, and computational resources that evolve over time. Backtesting such models during periods before their widespread adoption may therefore overstate their economic value. Supporting this concern, Figure 1 illustrates that the performance of more complex architectures, particularly neural networks, deteriorates in the later part of the sample, coinciding with the broader adoption of modern deep learning tools such as the Adam optimiser (Kingma and Ba 2014). These trends suggest a temporary technological advantage that diminishes as these methods become more accessible, highlighting the need for a more systematic investigation of anachronistic testing-date bias.

We study the effect of a model public release on its performance using a sample of eight machine learning models estimated on a broad cross-section of US common equity stocks over the out-of-sample period from January 2005 to December 2024. Our feature set follows Azevedo et al. (2023) and comprises 325 anomaly signals identified by Chen and Zimmermann (2022), 49 industry dummy variables from Fama and French (1997), and eight macroeconomic predictors from Welch and Goyal (2008). We assign each model a public release date corresponding to the first public release of the underlying algorithm or architecture. We form value-weighted long-short

decile portfolios and measure performance both gross and net of transaction costs, using the low-frequency bid-ask spread estimator of Chen and Velikov (2023). To test whether performance declines after a model’s public release, we utilise three complementary identification strategies. We begin with a structural-break regression on returns, using a method similar to that of McLean and Pontiff (2016), which serves as a benchmark test of whether long-short returns decrease once a model becomes publicly available. Next, we examine two factor-adjusted models that focus on abnormal returns. A decline in returns following a public release does not necessarily mean that the model has become less predictive, as overall market conditions and common risk factors influence returns. During periods of market-wide stress, such as the Covid-19 pandemic, underperformance may be due to unfavourable aggregate conditions rather than a true decline in stock-selection ability. To address this issue, we propose two structural-break regression models to estimate abnormal returns. The first model estimates factor-adjusted performance separately for each approach, allowing both alpha and factor loadings to change after the release. The second model pools results across models and constrains the post-release alpha shift to be common across all models. This trade-off sacrifices some flexibility to achieve greater statistical precision in estimating the average decay. These methods help us determine whether performance declines still occur after accounting for market-wide effects.

Our main findings can be summarised in three key points. First, across the entire sample, nearly all models produced positive cumulative gross long-short returns, with feedforward neural networks and elastic net performing best, whereas gradient-boosted tree methods generally fall behind. Second, after accounting for transaction costs, none of the strategies yields a statistically significant positive net return. This finding directly parallels the evidence in Novy-Marx and Velikov (2016) and Chen and Velikov (2023), who document that much of the apparent profitability of cross-sectional strategies disappears when realistic implementation frictions are accounted for. Third, and most closely related to our central hypothesis, the post-release regressions indicate that

model performance tends to decline after public release. The strongest statistical evidence comes from the pooled McLean–Pontiff-style regression, which estimates an average post-release return decline of 0.512% per month, statistically significant at the 5% level. The pooled factor-adjusted regressions point in the same direction: the estimated common post-release alpha shift is -0.523% per month in the full specification that allows factor loadings to vary after release, and -0.622% in the restricted specification that holds them fixed, with the latter statistically significant at the 10% level. At the individual model level, none of the factor-adjusted estimates is statistically significant on its own. Nevertheless, the sign pattern remains consistent with post-release deterioration: six of the seven model-specific alpha shifts are negative. Figure 2 shows that the point estimates are mostly below zero, although the confidence intervals are still quite wide.

The findings have several significant economic and practical implications. The existence of anachronistic testing-date bias implies that the alpha estimates reported in the machine learning asset pricing literature overstate the returns an investor could have earned by adopting these strategies after their public release. This bias could be operating through two main channels. First, the pre-release period, during which the strategy was unknown to market participants, is included in the performance evaluation and could mechanically inflate the reported average alpha. Second, a full-sample backtest implicitly assumes that the abnormal performance observed before public release will persist once investors can implement the strategy. In contrast, our evidence suggests that post-release performance is lower and, in the factor-adjusted specifications, remains weaker even after controlling for systematic risk exposures. More broadly, our results are consistent with the efficient markets hypothesis operating with a delay: once a strategy’s methodology is disclosed in the academic literature, sophisticated market participants learn about it, allocate capital towards it, and erode the risk-adjusted returns until they no longer cover transaction costs. This pattern aligns with the publication-induced decay documented for simpler return anomalies by McLean and Pontiff (2016). Our findings suggest the same mechanism extends to the machine

learning strategies that have attracted substantial interest in recent years. For investors who evaluate model performance through backtesting, our results highlight the importance of using release-date-aware out-of-sample windows and discounting reported pre-release performance when assessing the forward-looking viability of a machine learning strategy.

Our paper builds upon four strands of the literature. The first is the rapidly growing literature on machine learning methods for predicting the cross-section of equity returns. Gu et al. (2020) provide the seminal study, evaluating a broad set of supervised learning algorithms on US equities and document that neural networks and gradient-boosted trees deliver the largest out-of-sample predictive gains relative to linear benchmarks. Gu et al. (2021) extend this framework by incorporating autoencoder models that jointly learn latent factors and characteristic-based loadings, thereby improving upon conditional factor models. Related work studies the structure of the predictor space and the mapping from characteristics to returns. Kozak et al. (2020) adopt a Bayesian shrinkage perspective and show that a small number of anomaly-based principal components well approximates the stochastic discount factor. Freyberger et al. (2020) use non-parametric group lasso methods to identify characteristics with independent predictive content, while Kelly et al. (2019) develop instrumented principal components analysis, which models factor loadings as functions of observable characteristics. Chen et al. (2024) develop a deep learning approach that jointly estimates the stochastic discount factor and its factor structure. More recently, Kelly et al. (2024) show that the benefits of model complexity increase with the size of the predictor set. Our paper complements this literature by identifying the temporal feasibility of the predictive technology as a source of overstated performance.

Our study is also related to the literature on trading frictions and the implementability of machine-learning strategies. Novy-Marx and Velikov (2016) study the after-trading-cost performance of anomalies and show that transaction costs are especially damaging to high-turnover

strategies. In contrast, lower-turnover strategies are more likely to remain profitable once investors account for frictions and employ cost-mitigation techniques. DeMiguel et al. (2020) show that transaction costs can increase the number of firm characteristics that are jointly relevant for an investor’s optimal portfolio, because combining characteristics often reduces implementation costs through offsetting trades. Chen and Velikov (2023) document that bid-ask spread costs are large enough to absorb most of the apparent profitability of high-turnover anomaly strategies. In a setting closer to ours, Azevedo et al. (2023) construct a backtesting framework for machine-learning strategies that explicitly accounts for transaction costs and post-publication decay, documenting a sharp reduction in net alpha once these frictions are incorporated. Detzel et al. (2023) compare factor models net of transaction costs and find that cost-adjusted performance is substantially lower than gross performance for many machine-learning-based strategies. More recently, Li et al. (2024b) show that, when trading costs are considered, different models may be optimal for spanning investors’ investment opportunities, depending on their level of absolute risk aversion. Our paper adds to this strand by showing that the apparent gross profitability of machine-learning strategies, already substantially reduced once transaction costs are accounted for, deteriorates further when the evaluation window is conditioned on each model’s public release date.

Our research is also aligned with the literature on post-publication decay, replication, and the reliability of published return predictors. The seminal paper by McLean and Pontiff (2016) provides the foundational evidence that return predictors lose approximately 32% of their in-sample predictive power following journal publication, attributing the decline to arbitrage capital flowing into the disclosed strategies. Harvey et al. (2016) show that the multiple testing problem implies that a large fraction of published cross-sectional predictors may be false discoveries, and Hou et al. (2020) find that a majority of published anomalies fail to replicate in extended samples. Jensen et al. (2023) offer a more optimistic assessment, documenting that most published anomalies replicate once multiple-testing adjustments are applied consistently. Green et al. (2017) document

that a relatively small number of firm characteristics provide genuinely independent predictive information, and Harvey and Liu (2021) show that the distribution of published Sharpe ratios is difficult to distinguish from selective reporting across a large pool of candidate models. Chen and Zimmermann (2022) complement this literature by cataloguing 325 cross-sectional anomalies in a unified open-source framework and find that some of them are not statistically significant out-of-sample. Our contribution to this strand is to extend the decay hypothesis from individual predictors to the predictive technology itself, showing that the algorithms used to generate predictions exhibit the same post-disclosure performance deterioration that McLean and Pontiff (2016) document for the signals they produce.

Finally, our paper relates to the market-efficiency literature and its implications for the arbitrage of newly disclosed strategies. Fama (1970, 1991) establish the theoretical benchmark: in a semi-strong efficient market, publicly disclosed sources of return predictability should be incorporated into prices once they become widely known. The limits-to-arbitrage literature qualifies this prediction. De Long et al. (1990) show that noise-trader risk can deter arbitrage, while Shleifer and Vishny (1997) emphasise the role of capital constraints, agency frictions, and transaction costs in allowing mispricing to persist. Gromb and Vayanos (2010) further formalise the effects of arbitrageurs' wealth and financing constraints on the speed of convergence to efficiency. To our knowledge, this paper is the first to estimate a structural break in risk-adjusted returns at each model's own public release date, thereby isolating the performance decay attributable to the temporal infeasibility of the predictive technology rather than the predictors it processes.

The remainder of this paper is organised as follows. Section 2 describes the machine learning models considered, along with their public release dates and hyperparameter tuning procedures. Section 3 presents the data, outlines the feature engineering pipeline, and details the methodology for portfolio construction and transaction costs. Section 4 reports the empirical results: we first

document gross long-short performance, then show how profitability erodes with transaction costs, and finally present the three public-release decay regression specifications. Section 5 concludes and discusses directions for future research.

2. Machine Learning Models

To ensure that our findings are directly comparable with recent evidence on the profitability of machine-learning (ML) strategies, we adopt the core model set and backtesting conventions of Azevedo et al. (2023) as our baseline. We then extend their framework to study how performance declines once models become practically feasible and publicly available.

We use one linear model: Elastic Net (ENET), introduced by Zou and Hastie (2005). While maintaining the simplicity and interpretability of a linear model, ENET is more robust to numerous correlated predictors, which is particularly relevant in our application. It augments the standard least-squares objective with a convex combination of the ℓ_1 and ℓ_2 penalties,

$$\min_{\beta} \left\{ \frac{1}{n} \|y - X\beta\|_2^2 + \lambda [\alpha \|\beta\|_1 + (1 - \alpha) \|\beta\|_2^2] \right\}, \quad (1)$$

where $\lambda > 0$ controls how much we penalise model complexity, and $\alpha \in [0, 1]$ governs the relative weight of each regularisation penalty. In particular, the ℓ_1 term encourages sparsity by shrinking the coefficients of uninformative features exactly to zero, while the ℓ_2 term stabilises the solution when predictors are highly correlated. Because Elastic Net was published in March 2005, its effective date lies very close to the start of our evaluation window, leaving only a very limited pre-release period in the sample.

We also implement three gradient-boosted tree models. The first is XGBoost (XGB), introduced by Chen and Guestrin (2016). Unlike random forests, which grow trees independently, XGBoost fits them sequentially, with each new tree improving on the residual errors of the existing

ensemble. XGBoost has shown strong performance in cross-sectional return prediction, as noted by Gu et al. (2020). The second is LightGBM (LGB), introduced by Ke et al. (2017). LGB uses a leaf-wise tree-growth strategy designed to improve computational efficiency while preserving the standard gradient-boosting objective. Finally, we implement CatBoost (CBT), introduced by Prokhorenkova et al. (2018). CBT combines ordered boosting with symmetric trees, which is intended to reduce prediction shift and improve generalisation in noisy tabular settings. For all three tree-based models, we tune the hyperparameters annually using the tree-structured Parzen estimator algorithm, which models the distribution of promising hyperparameter regions and concentrates evaluations there rather than sampling uniformly across the search space. Although this particular implementation relies on the Optuna library (Akiba et al. 2019), which was not yet available to investors at the start of our sample, it simply allows us to navigate the hyperparameter space more efficiently. An investor tuning their hyperparameters year after year could have identified similar configurations using a standard grid or random search, albeit at the expense of increased computation time.

Finally, we implement feedforward neural networks (FFNNs) with two, three, four, and five hidden layers, labelled FFNN2, FFNN3, FFNN4, and FFNN5, respectively. Each hidden layer uses a ReLU activation function, batch normalisation to stabilise gradient flows during training, and dropout regularisation. The increasing depth of these architectures allows the models to learn progressively more abstract and non-linear functions of the firm characteristics. All neural-network models are optimised using the Adam algorithm by Kingma and Ba (2014). We date their effective availability to December 2014, when Adam was first published, using this as a proxy for the practical availability of modern deep-learning methods.

A complete overview of the models and their public release dates is provided in Table 1. These dates should be viewed as proxies for when a model became practically deployable and

sufficiently recognised by market participants, rather than as precise measures of market adoption. Specifically, the diffusion of a forecasting technology is likely to occur gradually, reflecting the spread of software implementations, computational resources, and awareness among practitioners over time, rather than a sudden change on a specific date. Therefore, our regressions utilise the effective release date as a clear and economically motivated breakpoint. However, the estimated coefficients should be interpreted as average changes in performance following the release, rather than as evidence of an immediate adjustment.

3. Data and Methodologies

This section describes the data, feature construction, portfolio methodology, and empirical design. Unless otherwise stated, the datasets considered cover the period from March 1957 to December 2024, although the out-of-sample machine-learning evaluation window begins later. In line with common practice in the literature, we restrict attention to U.S. common stocks listed on NYSE, AMEX, and Nasdaq (CRSP exchange codes 1, 2, and 3, and CRSP share codes 10 and 11). As a result, our dataset contains around 3.5 million stock-month observations over nearly 70 years. Our objective is to predict the total return, including any delisting, at time $t + 1$ using data available up to time t . To mitigate results influenced by delisting bias, as first explored by Shumway (1997), we estimate missing performance-related delistings at -35% for stocks listed on NYSE and AMEX, and -55% for stocks listed on NASDAQ.

We examine a wide array of firm characteristics, following the approach of Azevedo et al. (2023). We consider the 325 anomalies identified by Chen and Zimmermann (2022). Although only 212 of these signals are known or are thought to have significant predictive power, we agree with Kelly et al. (2024) that it is advantageous to consider all available anomalies. This approach enables us to broaden our feature space, thereby enhancing the effectiveness of our machine learning models. The anomaly data are downloaded using the `openassetpricing` Python package by Li et al.

(2024a). To improve the performance of machine learning algorithms, we normalise all features by mapping them to percentile ranks between -1 and 1 and replace missing values with zero. This procedure not only maps all anomalies onto a common scale (and thus stabilises the ML algorithms), but also makes them less sensitive to extreme values in the data. To expand our feature dataset, we also consider fixed industrial and temporal effects. We used the 49 publicly available industry classifiers from Fama and French (1997) to capture industry relationships. In the empirical specifications, the industry component enters through 48 dummy variables, with one industry omitted as the reference category to avoid perfect multicollinearity. In addition, in accordance with Azevedo et al. (2023), we include eight macroeconomic indicators from Welch and Goyal (2008) to capture movements in the macroeconomic landscape. In particular, we are considering the dividend-price ratio, earnings-price ratio, book-to-market ratio, net equity expansion, the treasury-bill rate, the term spread, the default spread, and stock variance. To ensure that we utilise only the information a potential trader had at time t , we consider only anomalies published before time t . For this reason, the number of features available to our models does not remain constant over time, but it increases. For instance, there were 192 features available in 2005, and this count has grown year after year, reaching a total of $325 + 48 + 8 = 381$ firm characteristics.

To evaluate the performance of our models as trading strategies, we construct a long-short portfolio based on forecasts of stock returns for the following month. Stocks are sorted according to their predicted returns. The long-short portfolio takes a long position in the top decile (stocks with the highest predicted returns) and a short position in the bottom decile (stocks with the lowest predicted returns). Stock weights are determined by market equity. To prevent very small firms from distorting the decile construction, given their greater exposure to extreme returns and transaction costs, we use NYSE breakpoints. The portfolio is formed at month t using forecasts of returns for month $t + 1$, and is then held from t to $t + 1$. After the returns of month $t + 1$ are

observed, the portfolio is rebalanced using the forecasts available for month $t + 2$. The long-short portfolio is self-financing because the long and short legs each have weights that sum to one in absolute value. This means that its gross return in month $t + 1$ equals the return on the long leg minus the return on the short leg:

$$r_{LS,t+1}^{gross} = r_{long,t+1}^{gross} - r_{short,t+1}^{gross}.$$

To make the backtests as realistic as possible, we account for transaction costs generated by rebalancing. Following Chen and Velikov (2023), we use the low-frequency bid-ask spread estimator *bas*, and define monthly trading costs as

$$TC_t = \sum_i |w_{i,t}^{target} - w_{i,t}^{owned}| \cdot bas_{i,t},$$

where

$$w_{i,t}^{owned} \propto (1 + r_{i,t}) \cdot w_{i,t-1}^{target}.$$

This reflects the fact that the portfolio weights before rebalancing depend on both the weights chosen in the previous period and the contemporaneous realised returns of the assets held. Net returns are obtained by subtracting transaction costs from both legs:

$$r_{LS,t+1}^{net} = r_{LS,t+1}^{gross} - (TC_{long,t} + TC_{short,t}).$$

As Chen and Velikov (2023) document, the low-frequency estimator tends to overstate effective trading costs relative to high-frequency benchmarks computed from quote data. Therefore, our net return estimates are conservative, representing a lower bound on what a practitioner with access to intraday spread data could achieve. We focus exclusively on the long-short strategy throughout the analysis because it provides a self-financing zero-investment spread that cleanly isolates the information content of the predicted returns without level effects. The cumulative performance of the long-short portfolios is reported as the compounded value of the monthly spread returns. Figure 1 plots the cumulative gross and net performance of each strategy from January 2005 to

December 2024.

We explore the effects of public model releases using three regressions, all estimated directly on net long-short returns. Each specification uses returns as the dependent variable rather than relying on a generated series of realised alphas. This approach avoids the inferential complications that arise in two-step procedures, where alphas are first estimated in a time-series regression and then used as dependent variables in a second-stage break regression. Because those realised alphas are themselves estimated objects, second-stage standard errors and test statistics must account for first-stage estimation error, see Pagan (1984) and Murphy and Topel (1985). The first specification is in the spirit of McLean and Pontiff (2016)’s pooled return-shift regression, adapted to our setting by replacing the publication of anomalies with the public release of predictive technology and by focusing on a single economically meaningful break at that date. In particular, we consider the following equation:

$$r_{i,t} = \alpha_i + \beta POST_{i,t} + \varepsilon_{i,t}, \quad (2)$$

where $r_{i,t}$ represents the net return in month t of the value-weighted long-short portfolio constructed from the predictions of model i , α_i is the fixed effect for model i , $POST_{i,t}$ is equal to one if month t occurs after the public release of model i , and zero otherwise, and β is the pooled average post-release return shift constrained to be common across models. This specification is the direct analogue of McLean and Pontiff (2016) applied to predictive technology rather than individual anomalies. Standard errors are computed using Newey–West heteroskedasticity and autocorrelation-consistent (HAC) estimators to account for serial dependence in returns.

However, a decline in raw returns does not always mean that the model has become less predictive. Instead, it could indicate adverse overall market conditions or a shift in the strategy’s exposure to systematic risk. To address this issue, we develop a second specification that consists

of a model-specific, factor-adjusted alpha-break regression, which we estimate separately for each model i :

$$r_{i,t} = \alpha_i + \Delta\alpha_i POST_{i,t} + \beta_i' F_t + \Delta\beta_i'(POST_{i,t} \times F_t) + \varepsilon_{i,t}, \quad (3)$$

where F_t is the vector of Fama-French five factors plus momentum, and $POST_{i,t} \times F_t$ denotes the element-wise product of the post-period dummy with each factor return. The pre-release alpha is α_i , and the post-release alpha is $\alpha_i + \Delta\alpha_i$, which implies that $\Delta\alpha_i$ is the primary object of interest for each model. Indeed, it represents the difference in the abnormal returns for each model before and after its release. The interaction terms $\Delta\beta_i'(POST_{i,t} \times F_t)$ explicitly allow factor loadings to shift at the public release date, ensuring that $\Delta\hat{\alpha}_i$ is not contaminated by changes in systematic risk exposures. We also estimate a restricted version that imposes $\Delta\beta_i = 0$, which forces factor loadings to be constant.

The third specification is the pooled factor-adjusted post-release alpha-shift regression, which pools evidence across all models to obtain a single average alpha shift with factor controls:

$$r_{i,t} = \alpha_i + \Delta\alpha POST_{i,t} + \beta_i' F_t + \Delta\beta_i'(POST_{i,t} \times F_t) + u_{i,t}, \quad (4)$$

where the variables are defined as previously, with the only change being that $\Delta\alpha$ now represents the pooled average difference in abnormal returns. Although this regression model is less flexible, it provides a single estimate for the post-release decay and offers greater statistical power.

As a robustness check, we conduct a placebo test for the Elastic Net (ENET). Because ENET's actual publication date of March 2005 lies very close to the start of our sample, it contributes too few pre-release observations for a meaningful model-specific alpha-shift regression and is therefore excluded from the main specification. To verify that the negative post-release shifts we document for the other models are not merely capturing broad adverse time trends rather than genuine effects of public disclosure, we construct a placebo treatment series labelled PLACEBO. This

series reuses the original ENET return data but assigns a placebo treatment date of March 2015, which falls in the same broad period as the true public-release dates of the other models in our sample (FFNN: 2014, XGBoost: 2016, LightGBM: 2017, CatBoost: 2018). Because the underlying returns are identical and only the event date changes, any estimated alpha shift for PLACEBO reflects purely a time effect anchored around 2015 rather than a genuine model-release effect. If the negative estimates for the treated models were driven primarily by a general deterioration in market conditions in the post-2015 period, we would expect PLACEBO to exhibit a similarly negative alpha shift. Conversely, if post-release decay reflects the gradual arbitrage of publicly disclosed predictive technology, the placebo estimate should be close to zero or positive.

4. Results

We organise this section into two parts. First, sections 4.1 and 4.2 document the gross and net long-short performance of the machine-learning strategies across the entire sample. These results are closely related to existing literature on return predictability, transaction costs, and the feasibility of implementing anomaly-based strategies, thus serving as a benchmark for our empirical framework. In the second part, we then focus on the paper’s main contribution. In particular, sections 4.3–4.5 examine whether the strategies’ performance diminishes after a model becomes publicly available. This analysis tests whether the anachronistic testing-date bias influences some of the reported profitability of machine-learning strategies.

4.1. Gross Long-Short Performance

We begin by assessing the overall predictive performance of each machine learning model over the full out-of-sample period from January 2005 to December 2024 before accounting for transaction costs. Table 2 reports the mean monthly long-short gross and net returns together with annualised Sharpe ratios. As shown in the table, visually supported by Figure 1, most models yield positive cumulative gross returns, with feedforward neural networks, elastic net, and CatBoost being

statistically significant at the 1% level. This order is partly consistent with the findings of Gu et al. (2020), who document that non-linear models tend to outperform linear benchmarks in cross-sectional return prediction. We conjecture that the tree-ensemble boosting models perform worse than the others due to the training methods we use. When training and tuning models, we follow Azevedo et al. (2023) and use the Mean Squared Error (MSE) loss. However, portfolio results are based on the ranking of predictions, as we long the top decile and short the bottom one. For smooth models, minimising MSE often maintains the correct ordering quite effectively. In contrast, with tree-based models, this is not always the case. A tree can reduce MSE by modifying just a few leaves without improving the relative rankings near the decile cutoffs. Consequently, we may place the stocks in the wrong decile, which accounts for their poorer performance.

4.2. The Role of Transaction Costs

Incorporating transaction costs leads to a significant reduction in returns. As indicated in the third and fourth columns of Table 2, the decrease in net returns relative to gross returns is substantial and consistent across all models. None of the strategies produces a statistically significant positive net return, with t-statistics below one in absolute value for all models except XGB and LGB, the latter of which exhibits a statistically significant negative net return. The other tree-ensemble boosting models also display negative net returns, while the FFNN architectures and ENET remain slightly positive but statistically insignificant. This outcome highlights the necessity of accounting for implementation frictions when assessing the practical viability of machine learning strategies, as emphasised by Novy-Marx and Velikov (2016) and Chen and Velikov (2023). The economic impact of cost drag is considerable. The difference between gross and net returns suggests that monthly rebalancing costs consume a large portion of the apparent cross-sectional alpha, consistent with Chen and Velikov (2023)’s finding that bid-ask spreads alone explain most of the profitability shortfall for high-turnover long-short strategies. Notably, Chen and Velikov (2023) also report that the low-frequency estimator used here tends to overstate effective trading costs

relative to high-frequency benchmarks based on intraday quote data. As a result, the reported net returns represent a conservative lower bound: investors with access to real-time spread data would likely experience less severe cost drag. However, given the magnitude of the gross-to-net gap, the principal finding is unlikely to change. For investors evaluating these models, the practical implication is that any realistically implementable strategy must incorporate cost-mitigation measures, such as reducing trading frequency, smoothing signals, or imposing turnover constraints, before gross performance figures can be considered indicative of net investment value.

4.3. Pooled Return-Shift Regression

As discussed in the previous section, we first test whether the decline in performance is attributable to anachronistic testing-date bias using the pooled return-shift regression in Equation 2, which aggregates net long-short returns across all models and estimates a single common return shift after public release, without factor controls. Table 3 presents the results. The pooled estimate $\hat{\beta}$ is -0.512% per month and is statistically significant at the 5% level. This corresponds to an economically significant decline of roughly 6.1 percentage points in average yearly returns. This result indicates that, on average, machine-learning strategies generated lower net returns following the public release of the underlying algorithms. This specification serves as the simplest and most transparent test of our central hypothesis, because it does not adjust for risk and applies a uniform post-release shift across all models. For these reasons, it is directly comparable to the framework proposed by McLean and Pontiff (2016). This estimate should be viewed as a reduced-form benchmark rather than a fully factor-adjusted measure of return decay. Changes in raw returns may also reflect broader shifts in market conditions or risk exposures. Nonetheless, the negative and statistically significant pooled estimate provides the clearest evidence in this paper that machine-learning return premiums decline after public release, laying the groundwork for the more structured factor-adjusted regressions discussed in Sections 4.4 and 4.5.

4.4. Model-Specific Factor-Adjusted Alpha-Shift Regression

To isolate public release-induced alpha decay within each model and to prevent shifts in systematic risk exposures from contaminating the estimates, we use the per-model structural-break specification of Equation (3), which allows both the alpha and the factor loadings to shift at each model’s own public release date. Table 4 reports the results. The estimated post-release alpha shift $\Delta\hat{\alpha}_i$ is negative for every model, except for FFNN3, where the estimate is economically small and statistically insignificant. The same general pattern of signs is observed in both the full specification, which permits factor loadings to vary at the time of release, and the restricted version, which keeps them constant. This suggests that the negative estimates are not simply a result of changes in systematic risk exposures. Although the magnitude of $\Delta\hat{\alpha}_i$ varies among the different models, the point estimates are largely consistent with post-release decay trends. That said, none of the individual estimates reaches statistical significance at conventional levels, indicating that the evidence for each model is more suggestive than definitive. Figure 2 strengthens this interpretation by showing that six of the seven point estimates lie below zero. While the confidence intervals are wide and generally include zero, the predominance of negative estimates across models is itself consistent with a broad decline in factor-adjusted performance following public release.

As a robustness check, the final row of Table 4 (shown in red in Figure 2) assigns a placebo release date of March 2015 to the Elastic Net return series. This placebo specification yields a positive post-release alpha-shift of 0.151% per month, which is not statistically significant ($p = 0.705$). Consequently, the placebo result does not replicate the predominantly negative pattern observed among the genuinely later machine-learning models. This reduces the concern that the main post-release estimates are driven solely by a generic deterioration in long-short strategy performance toward the end of the sample. Instead, it supports the interpretation that the negative coefficients for the later models are related to the timing of their public availability, rather than reflecting a common time trend.

4.5. Pooled Factor-Adjusted Alpha-Shift Regression

A more parsimonious estimate of the average post-release alpha shift is obtained by pooling information across all models using the panel specification of Equation (4), which constrains $\Delta\alpha$ to be common across models while allowing heterogeneous factor loadings before and after release. Table 5 presents the results. The pooled estimate $\Delta\hat{\alpha}$ is negative in both model specifications. In the full model, which allows factor loadings to change at the time of release, the estimated post-release alpha shift is -0.523% per month, although it is not statistically significant at conventional levels ($p = 0.126$). In the restricted model, which assumes constant factor loadings, the estimated shift is -0.622% per month and is statistically significant at the 10% level ($p = 0.063$). Importantly, both estimates are slightly larger in magnitude than the pooled return-shift estimate of -0.512% per month reported in Section 4.3. This comparison suggests that post-release decline is not primarily driven by exposure to systematic risk factors. In fact, the decline remains negative after accounting for factor exposures, with the restricted specification statistically significant at the 10% level. Additionally, the close proximity of the full and restricted estimates further indicates that changes in factor loadings are not the main reason for this result. By pooling models, this approach increases statistical power compared to analyses conducted on individual models, though it assumes uniform changes in performance across all architectures. Therefore, the pooled evidence adjusted for factors should be viewed as indicative of an average decline in abnormal performance following the public release. However, it is less statistically precise than the pooled return-shift results presented in Section 4.3. Collectively, the three specifications point to lower post-release performance, with the strongest statistical support coming from the pooled return-shift regression and weaker but directionally consistent evidence in the factor-adjusted specifications.

5. Conclusions

This paper studies whether the reported outperformance of machine learning models in predicting cross-sectional stock returns reflects genuine economic alpha or is partly inflated by anachronistic

testing-date bias, that is, the inclusion of pre-release periods in which markets had no opportunity to arbitrage the strategy. We analyse eight models over the out-of-sample period from January 2005 to December 2024 and evaluate their value-weighted long-short decile portfolios using three complementary post-release specifications.

Our main findings can be summarised as follows. First, across the full sample, nearly all models generate positive cumulative gross long-short returns, with Elastic Net and the feedforward neural networks performing best on a gross basis, while LightGBM is the only model with negative gross profitability. Second, once realistic transaction costs are incorporated, none of the models delivers a statistically significant positive net return, consistent with the evidence in Chen and Velikov (2023) that bid-ask spread costs absorb much of the apparent profitability of high-turnover strategies. Third, the strongest evidence of post-release decay comes from the pooled McLean–Pontiff-style return-shift regression, which estimates an average post-release decline of 0.512% per month, statistically significant at the 5% level. The pooled factor-adjusted regressions are directionally consistent with this result: they estimate post-release alpha shifts of -0.523% and -0.622% per month in the full and restricted specifications, respectively, with the latter significant at the 10% level. At the individual-model level, the factor-adjusted break regressions are not statistically significant, but six of the seven estimated post-release alpha shifts are negative, a pattern that is directionally consistent with post-release deterioration. A placebo robustness check further supports the causal interpretation. Assigning a synthetic March 2015 treatment date to the Elastic Net return series yields a positive estimated alpha-shift, in sharp contrast to the negative estimates obtained for models truly released around that period. This asymmetry is inconsistent with a purely time-trend explanation and supports the interpretation that public disclosure plays an important role.

Taken together, these results suggest that part of the machine-learning return premium documented in the literature reflects a temporary technological advantage that is gradually eroded once

the underlying methods become publicly available. Therefore, our findings warn against interpreting full-sample backtests of machine-learning strategies as evidence of persistent economic alpha, especially when these backtests include periods before the relevant predictive technology had been widely adopted. More broadly, the results support the idea that market participants learn about newly disclosed predictive methods and, over time, arbitrage away at least some of the abnormal performance. Future research could replace fixed release dates with a continuous diffusion index for machine learning methods. This index could be based on academic publications, software releases, code availability, and other indicators of model visibility and adoption. Such an approach would enable us to model post-release decay as a gradual process rather than a sudden change. An additional area of exploration could involve incorporating measures of computational feasibility, such as performance metrics for leading GPUs and CPUs. Investigating how these changes interact with model complexity could help us understand whether improvements in computing power accelerate the adoption of, and the decline in performance of, more resource-intensive machine learning architectures.

References

- T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama. Optuna: A next-generation hyperparameter optimization framework. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 2019.
- V. Azevedo, C. Hoegner, and M. Velikov. The expected returns on machine-learning strategies. Working paper, November 2023.
- A. Y. Chen and M. Velikov. Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, 58(3):968–1004, 2023.
- A. Y. Chen and T. Zimmermann. Open source cross-sectional asset pricing. *Critical Review of Finance*, 27(2):207–264, 2022.
- L. Chen, M. Pelger, and J. Zhu. Deep learning in asset pricing. *Management Science*, 70(2):714–750, 2024.
- T. Chen and C. Guestrin. Xgboost: A scalable tree boosting system. *arXiv preprint arXiv:1603.02754*, March 2016.
- J. B. De Long, A. Shleifer, L. H. Summers, and R. J. Waldmann. Noise trader risk in financial markets. *Journal of Political Economy*, 98(4):703–738, 1990.
- V. DeMiguel, A. Martin-Utrera, F. J. Nogales, and R. Uppal. A transaction-cost perspective on the multitude of firm characteristics. *Review of Financial Studies*, 33(5):2180–2222, 2020.
- A. Detzel, R. Novy-Marx, and M. Velikov. Model comparison with transaction costs. *The Journal of Finance*, 78(3):1743–1775, 2023.
- E. F. Fama. Efficient capital markets: A review of theory and empirical work. *The Journal of Finance*, 25(2):383–417, 1970.
- E. F. Fama. Efficient capital markets: II. *The Journal of Finance*, 46(5):1575–1617, 1991.
- E. F. Fama and K. R. French. Industry costs of equity. *Journal of Financial Economics*, 43(2):153–193, 1997.
- J. Freyberger, A. Neuhierl, and M. Weber. Dissecting characteristics nonparametrically. *The Review of Financial Studies*, 33(5):2326–2377, 2020.

- J. Green, J. R. M. Hand, and X. F. Zhang. The characteristics that provide independent information about average U.S. monthly stock returns. *The Review of Financial Studies*, 30(12):4389–4436, 2017.
- D. Gromb and D. Vayanos. Limits of arbitrage: The state of the theory. *Annual Review of Financial Economics*, 2(1):251–275, 2010.
- S. Gu, B. Kelly, and D. Xiu. Empirical asset pricing via machine learning. *The Review of Financial Studies*, 33(5):2223–2273, 2020.
- S. Gu, B. Kelly, and D. Xiu. Autoencoder asset pricing models. *Journal of Econometrics*, 222(1):429–450, 2021.
- C. R. Harvey and Y. Liu. Lucky factors. *Journal of Financial Economics*, 141(2):413–435, 2021.
- C. R. Harvey, Y. Liu, and H. Zhu. ... and the cross-section of expected returns. *The Review of Financial Studies*, 29(1):5–68, 2016.
- K. Hou, C. Xue, and L. Zhang. Replicating anomalies. *The Review of Financial Studies*, 33(5):2019–2133, 2020.
- T. I. Jensen, B. T. Kelly, and L. H. Pedersen. Is there a replication crisis in finance? *The Journal of Finance*, 78(5):2465–2518, 2023.
- G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu. Lightgbm: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems*, 2017.
- B. Kelly, S. Malamud, and K. Zhou. The virtue of complexity in return prediction. *The Journal of Finance*, 79(1), 2024.
- B. T. Kelly, S. Pruitt, and Y. Su. Characteristics are covariances: A unified model of risk and return. *Journal of Financial Economics*, 134(3):501–524, 2019.
- D. P. Kingma and J. Ba. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, December 2014.
- S. Kozak, S. Nagel, and S. Santosh. Shrinking the cross section. *Journal of Financial Economics*, 135(2): 271–292, 2020.
- P. Li, A. Chen, and T. Zimmermann. openassetpricing: Open asset pricing in python, 2024a. Python package.

- S. Li, V. DeMiguel, and A. Martin-Utrera. Comparing factor models with price-impact costs. *Journal of Financial Economics*, 162:1–26, 2024b. Forthcoming.
- R. D. McLean and J. Pontiff. Does academic research destroy stock return predictability? *The Journal of Finance*, 71(1):5–32, 2016.
- K. M. Murphy and R. H. Topel. Estimation and inference in two-step econometric models. *Journal of Business & Economic Statistics*, 3(4):370–379, 1985.
- R. Novy-Marx and M. Velikov. A taxonomy of anomalies and their trading costs. *The Review of Financial Studies*, 29(1):104–147, 2016.
- A. Pagan. Econometric issues in the analysis of regressions with generated regressors. *International Economic Review*, 25(1):221–247, 1984.
- L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin. Catboost: Unbiased boosting with categorical features. In *Advances in Neural Information Processing Systems*, 2018.
- A. Shleifer and R. W. Vishny. The limits of arbitrage. *The Journal of Finance*, 52(1):35–55, 1997.
- T. Shumway. The delisting bias in crsp data. *The Journal of Finance*, 52(1):327–340, 1997. ISSN 00221082, 15406261.
- I. Welch and A. Goyal. A comprehensive look at the empirical performance of equity premium prediction. *The Review of Financial Studies*, 21(4):1455–1508, 2008.
- H. Zou and T. Hastie. Regularization and variable selection via the elastic net. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 67(2):301–320, 03 2005. ISSN 1369-7412.

Figure 1 Cumulative gross and net performance of machine learning long-short portfolios

This figure presents the cumulative gross and net returns of a value-weighted long-short portfolio (order stocks by predicted returns, long the top decile, short the bottom decile) for eight machine learning models from January 2005 to December 2024. Portfolios are formed monthly on US common shares listed on NYSE, AMEX, and Nasdaq, using NYSE decile breakpoints. In the bottom chart, returns are net of transaction costs and are computed using the low-frequency bid-ask spread estimator of Chen and Velikov (2023). The y-axis uses a logarithmic scale, with the models' maximum and minimum final bank values indicated on the right-hand side.

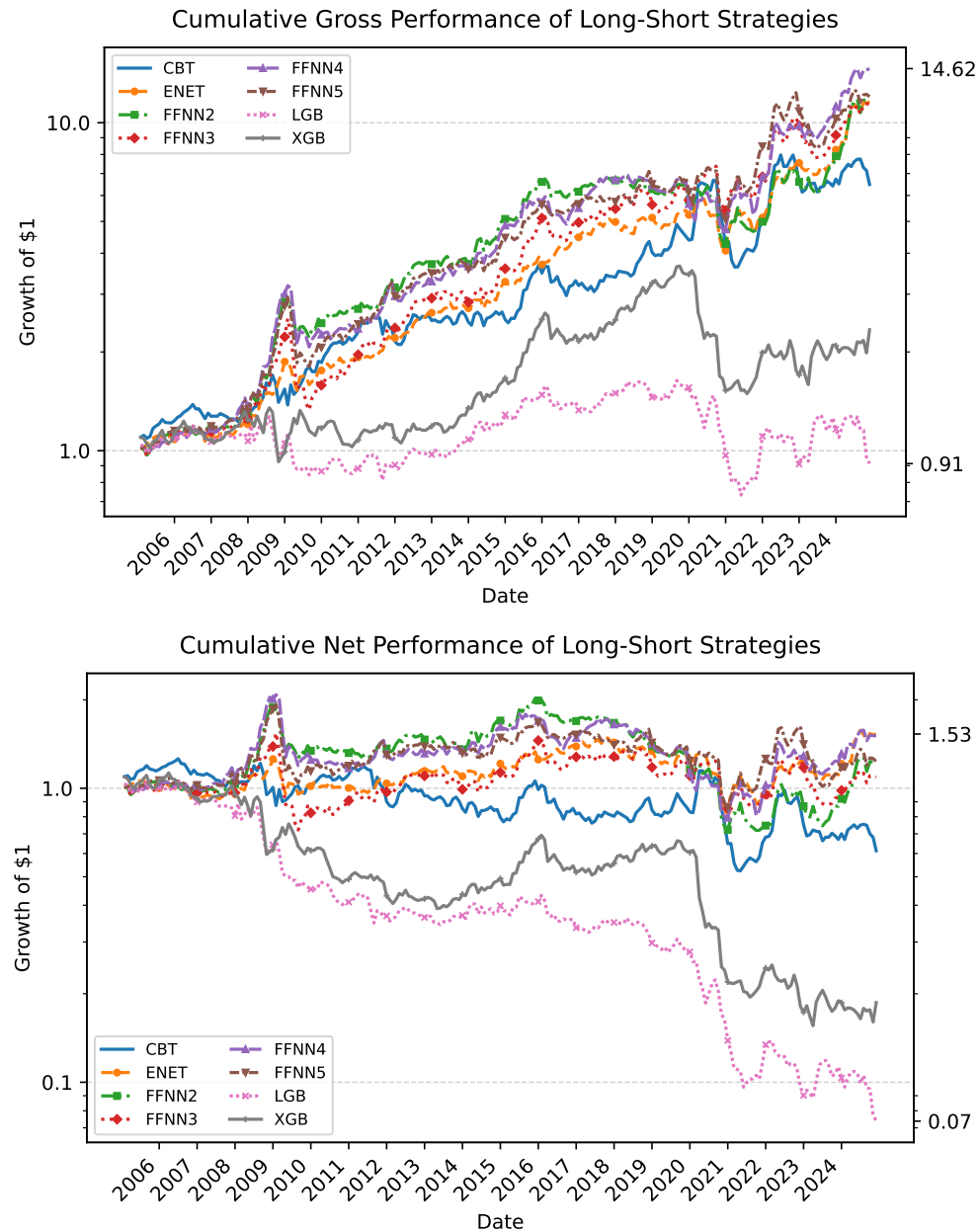


Figure 2 Model-Specific Factor-Adjusted Alpha-Shift

This figure reports the point estimate of $\Delta\hat{\alpha}_i$ for each model from the model-specific structural-break regression in Table 4. Dots denote the estimated post-release alpha shift, thick horizontal lines denote 90% confidence intervals, and thin horizontal lines denote 95% confidence intervals, all computed using Newey–West HAC standard errors. The dashed vertical line marks zero. Negative values indicate lower factor-adjusted long-short returns after the model’s public release. The entry labelled PLACEBO (shown in red) assigns a placebo March 2015 treatment date to the Elastic Net return series as a robustness check.

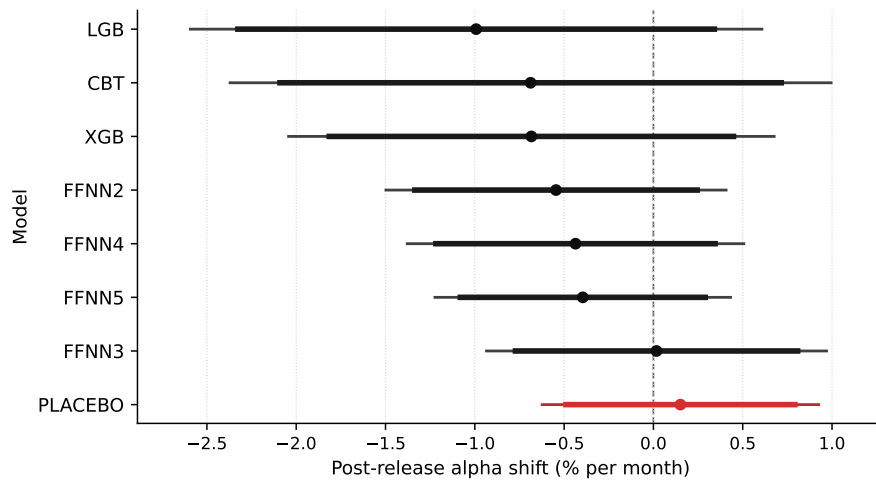


Table 1 Machine Learning Models: Public Release Dates

This table lists the release dates for different machine learning models. For each model, *Released* is the date on which the algorithm first appeared in the academic or technical literature. *Effective date* is the date from which the model is treated as practically deployable by investors. It determines the *POST* dummy in all regressions ($POST_{i,t} = 1$ for all months t on or after the effective date of model i). For the feedforward neural network architectures (FFNN2–FFNN5), the released date refers to the original neural-network literature (1980s), but the effective date is December 2014, the date of the Adam optimiser (Kingma and Ba 2014), which we use as a proxy for the practical availability of modern deep learning methods.

Code	Model name	Model class	Reference	Released	Effective date
ENET	Elastic Net	Linear regularised	Zou and Hastie (2005)	Mar 2005	Mar 2005
XGB	XGBoost	Tree ensemble (boosting)	Chen and Guestrin (2016)	Mar 2016	Mar 2016
LGB	LightGBM	Tree ensemble (boosting)	Ke et al. (2017)	Dec 2017	Dec 2017
CBT	CatBoost	Tree ensemble (boosting)	Prokhorenkova et al. (2018)	Dec 2018	Dec 2018
FFNN	Feedforward Neural Networks	Deep neural network	Kingma and Ba (2014)	1980s	Dec 2014

Table 2 Long-Short Portfolio Returns

This table reports the mean monthly gross and net returns for each machine-learning long-short strategy over the full sample period from January 2005 to December 2024. Portfolios are value-weighted, taking a long position in the top return-predicted decile (D10) and a short position in the bottom decile (D1), constructed using NYSE breakpoints on US common equities listed on NYSE, AMEX, and Nasdaq (CRSP share codes 10 and 11). Net returns subtract the low-frequency bid-ask spread cost estimator of Chen and Velikov (2023); gross returns are computed before any cost adjustment. SR denotes the annualised Sharpe ratio, computed as $\sqrt{12}$ times the ratio of the mean to the standard deviation of monthly returns, and TO% is the mean monthly two-way average-leg turnover. *t*-statistics in brackets are based on Newey–West HAC standard errors. Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (two-tailed).

Model	Gross			Net			TO%
	Ret%	[t]	SR	Ret%	[t]	SR	
FFNN4	1.263***	[3.75]	0.84	0.307	[0.92]	0.21	130.4
FFNN5	1.179***	[3.54]	0.79	0.222	[0.67]	0.15	131.7
FFNN2	1.163***	[3.66]	0.82	0.211	[0.67]	0.15	130.9
FFNN3	1.153***	[3.50]	0.79	0.169	[0.51]	0.11	134.0
ENET	1.102***	[4.46]	1.00	0.249	[1.01]	0.23	142.2
CBT	0.919***	[2.72]	0.61	-0.066	[-0.20]	-0.04	131.8
XGB	0.504	[1.45]	0.32	-0.541	[-1.52]	-0.34	144.7
LGB	0.089	[0.27]	0.06	-0.951***	[-2.87]	-0.64	145.4

Table 3 McLean–Pontiff Panel Regression: Post-Release Return Shift

This table reports estimates of Equation (2), a pooled panel OLS regression of net monthly long-short returns on a post-release indicator, with model fixed effects and no factor controls. $POST_{i,t}$ equals one from each model’s public release date onward, and the coefficient $\hat{\beta}_{POST}$ captures the average shift in net returns following public disclosure; a negative estimate indicates lower returns after release. Returns are net of transaction costs using the low-frequency bid-ask spread estimator of Chen and Velikov 2023. Standard errors are Newey–West HAC. Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (two-tailed).

	$\hat{\beta}_{POST}\%$	[t]	p-val	N	R^2
$\hat{\beta}_{POST}$	-0.512**	[-2.05]	0.040	1912	0.009

Table 4 Per-Model Post Public Release Alpha Shift

This table reports structural-break estimates of Equation (3), estimated separately for each model via OLS with Newey–West HAC standard errors over the period from January 2005 to December 2024. The regression allows both the intercept and the Fama–French five-factor-plus-momentum loadings to shift at each model’s public release date. $\hat{\alpha}_{pre}$ is the pre-release intercept and $\hat{\alpha}_{post} = \hat{\alpha}_i + \Delta\hat{\alpha}_i$ is its post-release counterpart; a negative $\Delta\hat{\alpha}_i$ indicates lower risk-adjusted performance after public disclosure. The column $p(\Delta\hat{\beta} = 0)$ reports the Wald-test p-value for joint stability of factor loadings at the release date. Returns are net of transaction costs following Chen and Velikov 2023. ENET is excluded from the main specification because its effective release date leaves too few pre-release observations in the sample. The row labelled PLACEBO is a falsification exercise that reuses the ENET return series but assigns it a placebo treatment date of March 2015. Stars on $\Delta\hat{\alpha}$: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (two-tailed).

Model	$\Delta\hat{\alpha}$ (POST coeff.)							
	$\hat{\alpha}_{pre}\%$	$\hat{\alpha}_{post}\%$	$\Delta\hat{\alpha}\%$	[t]	p-val	$p(\Delta\hat{\beta} = 0)$	N-pre	N-post
CBT	-0.088	-0.776	-0.688	[-0.80]	0.425	0.002	166	73
FFNN2	0.257	-0.289	-0.545	[-1.11]	0.266	0.500	118	121
FFNN3	-0.124	-0.107	0.017	[0.04]	0.972	0.398	118	121
FFNN4	0.387	-0.049	-0.436	[-0.90]	0.368	0.001	118	121
FFNN5	0.170	-0.225	-0.396	[-0.93]	0.353	0.280	118	121
LGB	-0.693	-1.686	-0.993	[-1.21]	0.226	0.297	154	85
XGB	-0.205	-0.889	-0.683	[-0.98]	0.327	0.305	133	106
PLACEBO	0.106	0.257	0.151	[0.38]	0.705	0.166	121	118

Table 5 Pooled Panel Regression: Post-Public Release Alpha Shift

This table reports estimates of Equation (4), a factor-controlled pooled panel that constrains the post-release alpha shift $\Delta\alpha$ to be common across models while allowing heterogeneous factor loadings before and after public release, over the period from January 2005 to December 2024. The restricted specification imposes constant factor loadings by omitting the $POST_{i,t} \times F_t$ interaction ($\lambda_i = 0$), and serves as a parsimony check on whether the results are driven by changes in systematic risk exposures. A negative $\Delta\hat{\alpha}$ indicates lower factor-adjusted returns following public disclosure. Standard errors are Newey–West HAC on the pooled panel, sorted by calendar date. Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (two-tailed).

Specification	$\Delta\hat{\alpha}\%$	[t]	p-value	N	R^2
Full (allows $\Delta\beta$)	-0.523	[-1.53]	0.126	1912	0.301
Restricted ($\Delta\beta = 0$)	-0.622*	[-1.86]	0.063	1912	0.257